

$$\begin{aligned} & \frac{1}{16\pi^2} \left(\frac{3}{4} m_1 \tilde{\mu} g_1^6 \text{LF}_{3,3,-2}[m_1, \tilde{\mu}] + \frac{1}{4} \tilde{\mu} g_1^6 m_1^3 \text{LF}_{3,3,-1}[m_1, \tilde{\mu}] + \frac{1}{6} m_2 \tilde{\mu} g_2^4 (g_1^2 + g_2^2) \text{LF}_{2,2,0}[m_2, \tilde{\mu}] - \right. \\ & \frac{1}{12} m_2 \tilde{\mu} g_2^4 (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[m_2, \tilde{\mu}] + \frac{11}{4} m_2 \tilde{\mu} g_2^6 \text{LF}_{3,3,-2}[m_2, \tilde{\mu}] + \\ & \frac{1}{4} \tilde{\mu} g_2^6 m_2^3 \text{LF}_{3,3,-1}[m_2, \tilde{\mu}] + \frac{1}{24} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} (g_1^2 + 3g_2^2) \text{LF}_{2,2,0}[m_d^r, m_q^p] + \\ & \frac{1}{12} \tilde{\mu} g_1^4 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \text{LF}_{3,1,0}[m_d^r, m_q^p] + \frac{1}{8} \tilde{\mu} g_1^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} (-g_1^2 + g_2^2) \text{LF}_{2,2,0}[m_e^r, m_l^p] + \\ & \frac{1}{4} \tilde{\mu} g_1^4 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \text{LF}_{3,1,0}[m_e^r, m_l^p] - \frac{1}{4} \tilde{\mu} g_1^2 g_2^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \text{LF}_{3,1,0}[m_l^p, m_e^r] + \\ & \frac{1}{16} \tilde{\mu} y_e^{\text{pr}} \overline{a_e^{\text{pr}}} (g_1^2 + g_2^2) (3g_1^2 + 5g_2^2) \text{LF}_{4,1,-1}[m_l^p, m_e^r] - \\ & \frac{1}{24} \tilde{\mu} y_e^{\text{pr}} \overline{a_e^{\text{pr}}} (g_1^2 + g_2^2) (3g_1^2 + 5g_2^2) \text{LF}_{5,1,-2}[m_l^p, m_e^r] - \\ & \frac{1}{12} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} (2g_1^2 + 3g_2^2) \text{LF}_{3,1,0}[m_q^p, m_d^r] + \frac{3}{16} \tilde{\mu} y_d^{\text{pr}} \overline{a_d^{\text{pr}}} (g_1^2 + g_2^2) (3g_1^2 + 5g_2^2) \\ & \text{LF}_{4,1,-1}[m_q^p, m_d^r] - \frac{1}{8} \tilde{\mu} y_d^{\text{pr}} \overline{a_d^{\text{pr}}} (g_1^2 + g_2^2) (3g_1^2 + 5g_2^2) \text{LF}_{5,1,-2}[m_q^p, m_d^r] + \\ & \frac{1}{24} \tilde{\mu} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} (-2g_1^4 + 9g_1^2 g_2^2 + 3g_2^4) \text{LF}_{2,2,0}[m_q^p, m_u^r] + \\ & \frac{1}{48} \tilde{\mu} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} (g_1^4 - 24g_1^2 g_2^2 - 9g_2^4) \text{LF}_{3,1,0}[m_q^p, m_u^r] + \\ & \frac{1}{8} \tilde{\mu} g_2^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[m_q^p, m_u^r] + \frac{3}{8} \tilde{\mu} g_2^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} (g_1^2 + g_2^2) \text{LF}_{4,1,-1}[m_q^p, m_u^r] + \\ & \frac{1}{48} \tilde{\mu} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} (7g_1^4 - 24g_1^2 g_2^2 - 15g_2^4) \text{LF}_{3,1,0}[m_u^r, m_q^p] - \\ & \frac{1}{8} \tilde{\mu} g_2^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[m_u^r, m_q^p] + \\ & \frac{3}{16} \tilde{\mu} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} (g_1^2 + g_2^2) (3g_1^2 + 5g_2^2) \text{LF}_{4,1,-1}[m_u^r, m_q^p] - \\ & \frac{1}{8} \tilde{\mu} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} (g_1^2 + g_2^2) (3g_1^2 + 5g_2^2) \text{LF}_{5,1,-2}[m_u^r, m_q^p] - \\ & \frac{1}{16} m_1 \tilde{\mu} g_1^2 \text{LF}_{3,1,0}[\tilde{\mu}, m_1] (g_1^2 + g_2^2)^2 - \frac{3}{8} m_1 \tilde{\mu} g_1^4 (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[\tilde{\mu}, m_1] + \\ & \frac{1}{16} m_1 \tilde{\mu} g_1^2 (g_1^2 + g_2^2) (3g_1^2 + 4g_2^2) \text{LF}_{4,1,-1}[\tilde{\mu}, m_1] + \\ & \frac{1}{4} m_1 \tilde{\mu} g_1^4 (g_1^2 + g_2^2) \text{LF}_{4,2,-2}[\tilde{\mu}, m_1] - \frac{1}{24} m_1 \tilde{\mu} g_1^2 (g_1^2 + g_2^2) (3g_1^2 + 5g_2^2) \text{LF}_{5,1,-2}[\tilde{\mu}, m_1] - \\ & \frac{1}{48} m_2 \tilde{\mu} g_2^2 (g_1^2 + g_2^2) (9g_1^2 + 17g_2^2) \text{LF}_{3,1,0}[\tilde{\mu}, m_2] - \frac{31}{24} m_2 \tilde{\mu} g_2^4 (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[\tilde{\mu}, m_2] + \\ & \frac{1}{16} m_2 \tilde{\mu} g_2^2 (g_1^2 + g_2^2) (9g_1^2 + 16g_2^2) \text{LF}_{4,1,-1}[\tilde{\mu}, m_2] + \\ & m_2 \tilde{\mu} g_2^4 (g_1^2 + g_2^2) \text{LF}_{4,2,-2}[\tilde{\mu}, m_2] - \frac{1}{8} m_2 \tilde{\mu} g_2^2 (g_1^2 + g_2^2) (3g_1^2 + 5g_2^2) \text{LF}_{5,1,-2}[\tilde{\mu}, m_2] - \\ & \frac{1}{8} \tilde{\mu} g_1^2 g_2^2 (g_1^2 + g_2^2) (7m_1 + 5m_2) \text{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] + \\ & \frac{1}{2} \tilde{\mu} g_1^2 g_2^2 (g_1^2 + g_2^2) (m_1 + m_2) \text{LF}_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] - \\ & \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \text{LF}_{2,1,1,0}[m_d^r, m_d^t, m_q^p] - \\ & \frac{3}{8} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} (g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_d^r, m_q^p, m_d^t] - \\ & \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \text{LF}_{2,1,1,0}[m_d^r, m_q^p, m_q^s] - \\ & \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \text{LF}_{2,1,1,0}[m_d^t, m_d^r, m_q^p] + \\ & \frac{3}{8} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} (g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_d^t, m_q^p, m_d^r] - \\ & \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \text{LF}_{2,1,1,0}[m_e^r, m_e^t, m_l^p] - \\ & \frac{1}{8} \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} (g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_e^r, m_l^p, m_e^t] - \\ & \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \text{LF}_{2,1,1,0}[m_e^r, m_l^p, m_l^s] - \\ & \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \text{LF}_{2,1,1,0}[m_e^t, m_e^r, m_l^p] + \\ & \frac{1}{8} \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} (g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_e^t, m_l^p, m_e^r] + \\ & \frac{1}{4} \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} (g_1^2 - g_2^2) \text{LF}_{2,1,1,0}[m_l^p, m_e^r, m_e^t] - \\ & \frac{1}{2} \tilde{\mu} g_2^2 \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \text{LF}_{2,1,1,0}[m_l^p, m_e^r, m_l^s] + \\ & \frac{1}{4} \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} (g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_l^p, m_e^r, m_l^s] - \\ & \frac{1}{8} \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} (g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_l^p, m_l^s, m_e^r] + \\ & \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \text{LF}_{2,1,1,0}[m_l^s, m_e^r, m_l^p] - \\ & \frac{1}{4} \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} (g_1^2 + g_2^2) \text{LF$$